



Shipping Route Optimization Analysis

Nassau Candy Distributor

Data Sources :

Project Field : Data Analytics

Project Name : Nassau Candy Distributor 

Tools Used : Python, Pandas, Numpy, Matplotlib, Seaborn, Plotly, Streamlit

Data Source : Nassau Candy Distributor.csv

Abstract :

In today's world, shipping products on time is very important for supply chain systems. It plays an important role in customer satisfaction and helps in managing the operational performance for the company to work well. This data analytics project emphasises on comprehensive data analytics study of Nassau Candy Distributor having 10k+ records. The analysis examines factory-to-customer route efficiency, lead time distributions, shipment delay rates, geographic bottlenecks, ship mode performance, and gross profit margins. The shipping routes help to find out at which places products are not shipped on time, why there is a delay and why there are delays in the logistic.

The project is made from tools like Python, Pandas, NumPy, Plotly, Matplotlib, Seaborn and Streamlit. The final streamlit web dashboard that helps users to explore route performance, geographic delay patterns, ship mode efficiency, Slowest routes, fastest routes, sales by region. These findings give important insights on how to make logistics better, reduce delays, make customers satisfied and happier.

Introduction :

Good logistics management is necessary for companies that ship things to different

regions. Nassau Candy Distributor ships products from factories to customers in different areas states and uses different ship modes. So it's very important to make sure that the routes are good, products are shipped efficiently, with making great logistics decisions for different routes. This research addresses two core questions : First, which factories, routes, ship modes, and geographic regions represent the greatest sources of shipping delay and customer delivery risk? Second, how can an interactive analytical dashboard be designed to serve as a decision-support tool that helps logistics managers to identify these bottlenecks and act on them efficiently? This project gives us a way to look at shipping efficiency find problems compare ship modes and make better decisions using a Streamlit dashboard.

Problem Statement :

Currently, Company is facing challenges in shipping route efficiency, logistics management, delay prone regions, and operational bottlenecks. There are some problems that arise the questions like:

- Which shipping routes are efficient and which are not efficient?
- Which regions face a lot of delays?
- Which regions having high delivery delays?
- Which slowest routes need immediate logistics attention?
- How do different ship modes affect delivery?

Objectives :

We want to analyse data in a proper manner and get some valuable insights by which we can find :

- Which routes are the fastest and which ones are slowest?
- How the different regions get their products?
- Which shipping modes are the best?
- Calculate Shipping lead time to ship each order.
- Find the worst routes.
- find out that the delivery delayed in which areas,by which ship modes and in which factories.
- Find high-volume bottlenecks states/regions and routes that need to be resolved.

Building an attractive dashboard for business users so that the best decisions can be made to improve company's overall performance.

Dataset Overview :

We have a dataset of Nassau Candy Distributor of 10,194 rows and 18 columns.

Field	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date of order
Ship Date	Date of shipment
Ship Mode	Shipping method of order
Customer ID	Unique customer identifier
Country/Region	Country or region of customer
City	City of customer
State/Province	State/province of customer
Postal Code	Postal code / zip code of customer
Division	Product division
Region	Region of customer
Product ID	Unique product identifier
Product Name	Product long name
Sales	Total sales value of order
Units	Total units of order
Gross Profit	Gross profit of order
Cost	Cost to manufacture

Methodology :

1. Data Cleaning :

Data processing and data cleaning is a crucial method in any data analytics project. Data is cleaned through various steps like:

- Convert the date into standardize date format(Parse format)
- Standardize column names/ labels(field)
- Remove null values
- Handle missing values
- Remove duplicates
- Remove invalid records

2. Feature Engineering :

To analyse the given data properly, we have to make some features like :

- Lead Time : to calculate time to deliver the product (in days)
- Mapped products to factories : to find which candy was made in which factory
- Route fields : to find the shipping route

3. Aggregation :

Data is grouped by factory, state, ship mode, route, region and division. Rank the routes as top 10 fastest routes and top 10 slowest routes. Compare the ship modes as most efficient shipping mode.

4. Creating KPI (Key Performance Indicator) :

Calculate the Shipping Lead Time(delivery time taken), gross profit, delay frequency, route volume, profit margin %, profit & revenue contribution % and efficiency score.

- Shipping Lead Time : Delivery Time duration between order date and shipping date.
- Average Lead Time : Average delivery performance across the routes and ship modes.
- Route volume : The number of shipments of product handled by each route.
- Route Efficiency Score : Ranks the routes according to score using Lead Time Performance
- Gross Profit

5. Creating Dashboard :

Create an attractive and interactive dashboard using python libraries like- matplotlib,seaborn and plotly.

Exploratory Data Analysis :

1. Lead Time Analysis :

Analysing the Lead Time means time taken between Order date and Ship date to deliver the products from factory to different states / region. We can find some more features related to lead time like:

- Shipping Lead Time
- Average Shipping Lead Time
- Average Lead Time by Region
- Average Lead time by ship mode
- Average Shipping Lead Time by Route Region

2. Ship Mode Performance Analysis :

There are different shipping modes. We can conclude that Same Day is the fastest and Standard Class is the slowest.

Shipping Mode directly Affects Delivery Speed And Delay Frequency.

Delivery performance varies across the regions/states and different routes.

3. Region-wise Shipping Performance Analysis :

There are 4 regions given in the dataset out of which the Interior region is the fastest and the Gulf region is the slowest among them.

4. Route Efficiency Score Analysis :

The Route Efficiency Score is a way to measure that how much timely the product will be delivered and how much the route is efficient for shipping. The score goes from 0 to 1 with 1 being the best. The routes that do the best are the ones that come from the Secret Factory in Illinois and The Other Factory in Tennessee. These factories are in the middle of the US which makes it easier to get to all the states.

The routes that do the worst are the ones that go across the country like from Arizona to the northeastern states or from Georgia to the Pacific Coast.

Data Visualization :

By creating KPIs and EDA , now we will visualize the data by created feature such as :

- Distribution of Shipping Lead Time
- Average Lead Time by region
- Lead Time Distribution by region
- Lead Time Distribution by Ship mode
- Top 10 fastest routes
- Top 10 slowest routes
- Top 10 product by profit
- Bottleneck Analysis

Streamlit App Dashboard :

The dashboard is created using Streamlit app by uploading github project to give important findings and valuable insights.

Dashboard :

<https://j8svmjdenerwer7k7ybu2kmz.streamlit.app>

Findings / Insights :

By analyzing the given dataset, we get some important findings that can help us understand how Nassau Candy logistics operates. Some insights are

- **Route Performance Analysis** : Some routes give better route performance , having good and efficient logistics network that helps to deliver faster than other routes.
- **Bottleneck Analysis by Region** : Company is facing some certain challenges due to higher Lead Time in some regions / states, routes and logistics related issues. Prioritize the routes with poor shipping performance(high lead time).
- **High Lead Time** : Some regions having certain routes that lead to high

shipment volume and high lead time and delivery delays at that routes.

- **Ship Mode** : Faster ship modes reduce delivery time but may increase cost.

Recommendations :

- Prioritize routes with a lot of logistics problems and long lead times.
- Improve distribution strategies to reduce delays.
- Use faster ship modes for priority shipments.
- Monitor route efficiency regularly.
- Improve route planning using region, factory and ship mode insights.

Conclusion :

This project gives us an analysis of shipping route efficiency for Nassau Candy Distributor. By turning data into useful insights the company can fix inefficient routes reduce delays and improve customer satisfaction.

The Streamlit dashboard makes the analysis interactive and easy to use. It lets users look at route performance, geographic patterns, ship mode efficiency, and bottlenecks, in time. Overall the project shows how data analytics and dashboarding can improve supply chain decision-making and performance.

Project Links :

Github Repository : <https://github.com/Nivedita-1998/Nassau-Candy-Distributor-Shipping-Optimization-Dashboard>

Streamlit App Dashboard : <https://j8svmjdenwer7k7ybu2kmz.streamlit.app/>